

Frame of reference

A frame of reference is a co-ordinate system by which the motion of a body can be described.

There are two frames of reference:

- (i) Inertial frame of reference
- (ii) Non-inertial frame of reference.

Inertial frame of reference: An inertial frame of reference is one in which *Newton's first law of motion is true.*

It may be stated in another way as, "*The frames with respect to which an unaccelerated body appears unaccelerated are inertial frames.*"

It another words, "*The frames which are at rest or in uniform translatory motion relative to one another are inertial frames.*"

Non-inertial frame of reference: A non-inertial frame of reference is one in which *Newton's first law of motion is not true.*

It may be stated in another way as, "*The frames with respect to which an unaccelerated body appears accelerated are non-inertial frames.*"

Relativity

It is a theory of physics that says, *all motion must be defined relative to a frame of reference and that space and times are relative.* This theory says that, *measurement of space and time are affected by the relative motion between an observer and what is observed.*

The theory of relativity is divided into two parts. These are:

- (i) Special theory of relativity.
- (ii) General theory of relativity.

Special theory of relativity: The natural phenomenon happening in inertial frame of reference is explained by means of special theory of relativity.

General theory of relativity: The natural phenomenon happening in non-inertial frame of reference is explained by means of general theory of relativity.

Postulates of special theory of relativity

The special theory of relativity is based upon two postulates:

(i) The principle of relativity: *The laws of physics are the same in all inertial frames of reference.*

Explanation: If the laws of physics were different for different observers in relative motion, the observer would determine from these differences which of them were stationary in space and which were moving. But such a distinction does not exist and the principle of relativity expresses this.

(ii) The principle of invariance of light: *The speed of light in free space has the same value in all inertial frames of reference.*

Explanation: According to this postulate the speed of light is same in all directions. No matter whether the source of light is moving or stationary.

Galilean transformation

Sometimes it is necessary to transform the co-ordinates from one reference frame to other.

The transformation of co-ordinates of a particle from one inertial frame of reference to another inertial frame of reference is called the Galilean transformation.

Let us consider two co-ordinate systems, one inertial system S and another S' moving uniformly with respect to S with velocity v . The geometric representation of the above system is given below:

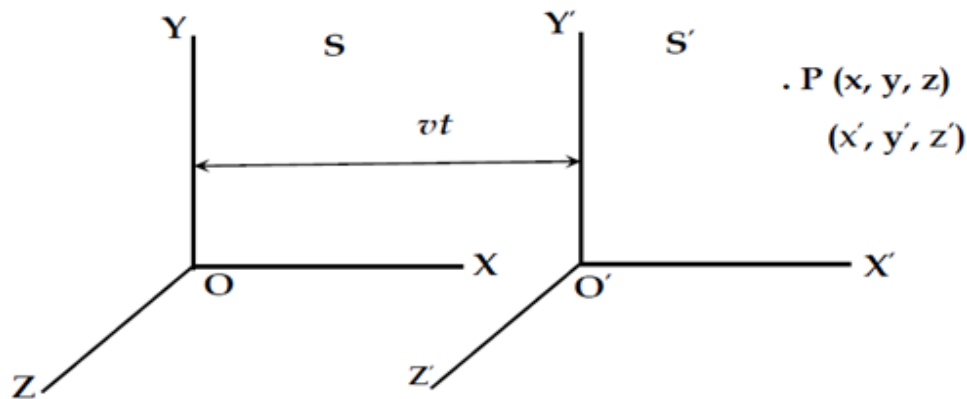


Fig. 1.

If clocks in the both systems are started when the origins of S and S' coincide, measurements in x -direction made in S will be greater than those made in S' by the amount vt , which is the distance S' has moved in the x -direction. *i.e.*

$$x' = x - vt \quad (1)$$

There is no relative motion in y - and z -directions and so

$$y' = y \quad (2)$$

$$z' = z \quad (3)$$

It is assumed that both systems are using the same time. It means that time measurements are independent of the motion.

$$t' = t \quad (4)$$

The set of equations (1), (2), (3) and (4) are called Galilean transformation equations.

To convert velocity component measured in S frame to their equivalent in the S' frame according to Galilean transformation, we have to differentiate simply x' , y' and z' with respect to t . Thus

$$v'_x = \frac{dx'}{dt} = v_x - v \quad (5)$$

$$v'_y = v_y \quad (6)$$

$$v'_z = v_z \quad (7)$$

Invariance of Newton's law of motion

The force acting on a particle as observed in frame S is given by

$$F = \frac{d}{dt}(Mv)$$

$$F = M \frac{dv}{dt}$$

$$\therefore F = Ma \quad (1)$$

The force acting on the particle as observed in frame S' is given by

$$F' = \frac{d}{dt}(Mv')$$

$$F' = M \frac{dv'}{dt}$$

$$\therefore F' = Ma' \quad (2)$$

According to Galilean transformation

$$v'_x = v_x - v$$

$$\frac{dx'}{dt} = \frac{dx}{dt} - v$$

Differentiating the above equation w. r. to t we get

$$\frac{d^2x'}{dt^2} = \frac{d^2x}{dt^2}$$

$$\therefore a' = a$$

From equation (1) and (2) we can write

$$F' = F$$

It means same force will be observed in two inertial frames of reference S and S' . It means force is invariant in Galilean transformation. It also means that Newton's second law of motion is valid in all inertial frames of reference. *i.e.* the basic laws of physics are invariant in two inertial frames of reference.

Lorentz transformation

The Lorentz transformation equations are

$$x' = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (1)$$

$$y' = y \quad (2)$$

$$z' = z \quad (3)$$

$$t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (4)$$

The inverse Lorentz transformation equations are

$$x = \frac{x' + vt}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (5)$$

$$y' = y \quad (6)$$

$$z' = z \quad (7)$$

$$t = \frac{t' + \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (8)$$

Length contraction

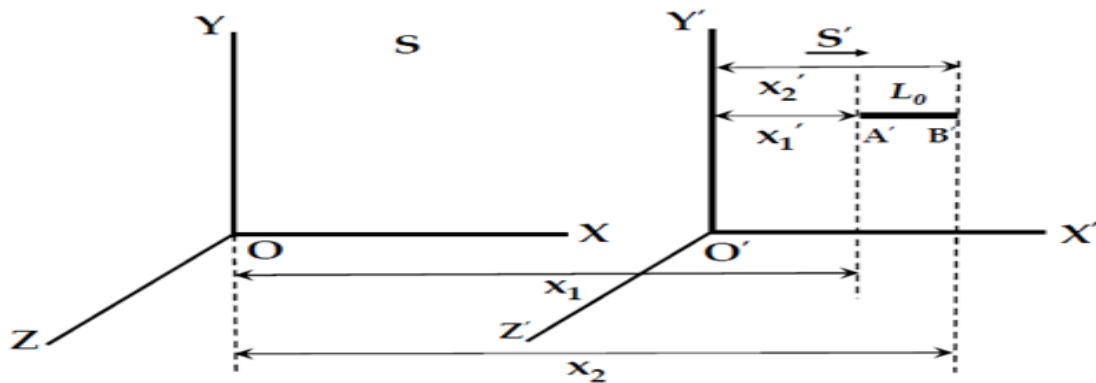


Fig. 2.

Consider the inertial systems S and S' . The system S' is moving to the right along x -axis with a uniform velocity v with respect to S . The object $A'B'$ (rod) lying parallel to its x' -axis. The coordinates of its ends are x'_1 and x'_2 .

If L_0 the proper length of the rod then the length of the rod measured by an observer in the frame S' .

$$L_0 = x'_2 - x'_1 \quad (1)$$

If x_1 and x_2 be the coordinates of the ends of the same rod in the frame S , then the length of the object lying in the moving frame S' as measured by S .

$$L = x_2 - x_1 \quad (2)$$

According to the Lorentz transformation equation

$$x'_1 = \frac{x_1 - vt}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$x'_2 = \frac{x_2 - vt}{\sqrt{1 - \frac{v^2}{c^2}}}$$

We can write

$$x'_2 - x'_1 = \frac{x_2 - x_1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Here,

$L_0 = x'_2 - x'_1$ is the actual or proper length of the rod.

$L = x_2 - x_1$ is the observed length of the same rod.

Thus we can write

$$L_0 = \frac{L}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\therefore L = L_0 \sqrt{1 - \frac{v^2}{c^2}} \quad (3)$$

This shows that the length of a rigid body in the direction of its motion with uniform velocity v is reduced by a factor $\sqrt{1 - \frac{v^2}{c^2}}$ in the direction of motion. This is called '*length contraction*'.

This also shows that *the length of a stationary object with respect to an observer in motion is shorter than the length measured by the observer at rest.*

When the velocity of the body is negligibly small as compared to light velocity c , then we can write

$$\frac{v}{c} = 0$$

then the contraction of length is negligible and

$$L = L_0$$

Thus when the velocity of the body is negligibly small as compared to light velocity c , the contraction of length is negligible and $L = L_0$.